

Brake Aid User Guide

Thank you for being part of Brake Aid's beta testing program.

Brake Aid provides players of Sim racing games such as *iRacing* and *Assetto Corsa*, a mechanism for fine tuning their braking performance, technique and timing, with the ultimate aim of reducing the player's lap times, as a result of incrementally tweaking their braking technique to get maximum performance from the simulated vehicle's braking system.

Contents

Overview of Brake Aid	2
Initial Benchmark Lap with no Brake Markers:	2
How Brake Aid's Brake Markers are determined:.....	2
How Brake Aid Displays the Dynamic Brake Markers:	2
Adjustment of Brake Markers to fine tune braking performance:	3
Preventing a Lap from Updating the Brake Markers:	4
Restricted Mode – Keeping a fixed set of brake points that don't update	4
Using the Software	6
Launching Brake Aid	6
Brake Aid Setup Menu	6
User Selection	6
Begin Session	6
Using Brake Aid	7
Configuring the Brake Aid Overlay Light Graphic	7
Exiting a Brake Aid Session	7
Button Mapping.....	8
Game Controller Compatibility	8
Regarding Keyboard Key Binding:	8
Mapping a Button/Key to a Brake Aid User Input Function:	8
Brake Aid Configuration	9
Troubleshooting	10
If the software crashes.....	10
Reporting Bugs	10
Third Party Notices	11

Overview of Brake Aid

Brake Aid is different from other driving aids that are based on a fixed idealised driving style, suited to the vehicle being driven and the nominal conditions of the racetrack being driven on.

Instead Brake Aid fully adapts itself to the player's driving style, by analysing their braking performance and technique as they approach and brake for each of the corners of a racetrack.

Initial Benchmark Lap with no Brake Markers:

Brake Aid bases its dynamic brake markers directly on the braking performance that it observes in the previous lap. As such, when the player begins a **new Brake Aid session** on a racetrack, the first lap driven becomes the initial "benchmark" lap for the system. During this lap, **Brake Aid provides no dynamic brake markers for any of the corners.**

On subsequent laps, or when Brake Aid is started in *Resume Last Session* mode, the system is then able to provide a dynamic brake marker for each of the corners that (in the prior lap) required *heavy* braking over a set intensity threshold.

How Brake Aid's Brake Markers are determined:

The brake markers provided by Brake Aid for each corner are *not* mere static points on the racetrack that correspond to the braking start position that the driver chose for that corner in the previous lap.

Instead Brake Aid's brake markers are fully dynamic and are dependent on real time vehicle speed and acceleration, as well as the driver's reaction time as they approach a corner.

When a Brake marker is signalled to the driver for a corner, it means that if the driver starts braking at this instant, and adopts similar braking intensity and technique/modulation as they did in the previous lap, **the vehicle will get to the same minimum speed within that corner, at the same position within that corner, as it did on the previous lap.**

Because the brake marker is fully dynamic and reaction time calibrated to the player, even if the player were to approach the corner 20% faster than on the previous lap, the brake marker provided by Brake Aid will still ensure that the player gets to the same minimum corner speed at the same position within the corner, as they did on the previous lap, although in this case, the player will observe that the brake marker is signalled noticeably earlier than on the previous lap, to account for the additional approach speed.

How Brake Aid Displays the Dynamic Brake Markers:

The dynamic brake markers provided by Brake Aid are communicated by Brake Aid's **Overlay light graphic** first illuminating in yellow **approximately one second before** the driver is due to reach the brake marker that the system has calculated.

This is known as the *anticipation light*, and its purpose is to prepare the driver to react and get ready to brake for the imminent brake marker.

The brake marker is then **signalled by the yellow light turning off**, which indicates the precise position where braking should start.

Adjustment of Brake Markers to fine tune braking performance:

In addition to the default mode of operation, where Brake Aid provides brake markers that allow the player to repeatably reach the same mid corner speed within each corner of a racetrack, Brake Aid offers the ability for the player to manually adjust the position of each brake marker on the track, *relative to* the default position that it calculates for the marker being adjusted (which would allow for the same minimum corner speed at the same position in the corner as was achieved on the previous lap).

This is achieved using Brake Aid's ***Push Forward*** and ***Pull Back*** inputs, which can be mapped to either a pair of keyboard keys or buttons on connected external peripherals/hardware.

During a lap, when the player uses a dynamic brake marker provided by Brake Aid for a corner, to assist them in braking for that corner, if the player feels that the brake marker came on *too early*, or they are able to navigate the corner at a faster speed, then after they finish braking for that corner, they can press the *Push Forward* input (mapped button/key) to shift the dynamic brake marker that will be displayed for that corner on next lap, closer towards the corner by increments of 2 metres per press of the *Push Forward* input.

Each time the *Push Forward* input is pressed, the overlay light graphic briefly flashes green to confirm that the press was registered by the system.

The *Push Forward* button can be pressed multiple times, each of which will shift the dynamic brake marker an additional 2 metres closer to the corner on the next lap.

If the player uses the *Push Forward* input for a corner's brake marker, when the player approaches that same corner on the next lap, they will be reminded of this change they made, by the one second anticipation light illuminating in **green** instead of yellow, before turning off to signal to the player to brake.

Conversely, if the player exits a corner that they braked *too late* for, they can tweak the dynamic brake marker that will be displayed for that corner on the next lap, by pressing the **Pull Back** input, which will force the dynamic brake marker to display earlier than the calculated default position, by increments of 2 metres.

Each time the *Pull Back* input is pressed, the overlay light graphic briefly flashes red to confirm that the press was registered by the system.

The *Pull Back* button can be pressed multiple times, each of which will shift the dynamic brake marker an additional 2 metres further away from the corner on the next lap.

If the player uses the *Pull Back* input for a corner's brake marker, when the player approaches that same corner on the next lap, they will be reminded of this change they made, by the one second anticipation light illuminating in **red** instead of yellow, before turning off to signal to the player to brake.

Preventing a Lap from Updating the Brake Markers:

The brake markers provided by Brake Aid for each of the corners on a racetrack are, by default, based on the braking performance observed by the system on the *immediate previous* lap. This is to allow the system to continually adapt to changing driving conditions, tyre wear, fuel loads, etc.

However, the player may drive a lap during which, on one of the corners, an incident occurs, such as encountering traffic, which may cause them to brake way too early, or switch to a suboptimal driving line, etc. Or the player may lose control or run off the track.

In these scenarios, this would likely result in poor or altered braking performance and technique that the driver doesn't want to repeat on subsequent laps.

Additionally, an incident on one of the corners may also reduce the vehicles exit speed from that corner, and consequently, reduce the entry speed to the next corner, which may also affect braking performance and technique through that corner.

Therefore, when a driving incident that affects braking performance for one of the corners occurs, the player has the option to *disregard* the entire lap (being driven) from updating the brake makers, therefore allowing the brake marker targets being used to calculate the dynamic brake markers on the current lap, to be carried over to the next lap also.

This function is achieved by pressing the **Don't Count** input at any point during the lap that the player wants to disregard.

When the **Don't Count** input is pressed, the overlay light graphic flashes red, yellow, then green in quick succession, before turning off, to confirm that the press was registered by the system, and the lap being driven will *not* be used to update the dynamic brake markers for the next lap.

Restricted Mode – Keeping a fixed set of brake points that don't update

In the scenario where a player has set an optimal lap with optimised braking points for all corners, that they don't want updated/adapted anymore on each lap, Brake Aid offers *restricted mode*, a feature which (when enabled) *prevents* the set of brake points currently being used by the system to drive the dynamic brake markers (at the time when restricted mode is turned on), *from being updated any further* on each lap.

Restricted mode is equivalent to the player manually pressing the *don't count* button on every lap going forward (until restricted mode is toggled off by the player).

Dynamic brake marker target speed adjustment function

In addition to preventing the current set of brake points from updating, *restricted mode* also offers the player the ability to automatically adjust all of the dynamic brake marker target mid corner speeds **by a single set factor**, adjustable in the UI's configuration settings page.

This feature is useful for *driver training purposes* such as *Pro Am racing* on a simulator, by, for example, a *pro driver/instructor* setting an optimal lap while using Brake Aid, then after this optimal lap, toggling restricted mode on, and if they choose, *dialling back* all of the dynamic brake marker target speeds by a certain factor/percentage (for example 80% of the *pro driver's* target mid corner speeds), via setting the restricted mode's target speed adjustment factor.

An *am driver* can then start training on the simulator using the (optionally) dialled back dynamic brake marker points previously set by the *pro driver/instructor*, without these optimal dynamic brake markers updating when the *am driver* completes a lap.

Toggling Restricted Mode On/Off

Restricted mode is toggled on or off by simply triple clicking the Brake Aid overlay graphic at any time during a race session.

If *restricted mode* has been *successfully toggled on*, the surround/border of the Brake Aid overlay graphic changes from **black** to **red**.

If restricted mode has been *toggled from on to off*, the surround/border of the Brake Aid overlay graphic changes from **red** to **black**.

Note: If the player starts a new *Brake Aid session*, *restricted mode* can only be toggled on if the player has driven *at least* one lap of the track and they have braked (heavy braking) for *at least* one corner on their previous lap.

Restricted mode can be toggled on immediately if *Resume Last Session* was used to recall the brake markers from a previous Brake Aid session.

Using the Software

Launching Brake Aid

In the Brake Aid Beta root directory, launch **Brake Aid Beta.exe** to open the Brake Aid user interface.

Brake Aid Setup Menu

The Brake Aid Setup menu allows the user to create or select a user profile, and then to start a Brake Aid *session* once a compatible game session has been found.

User Selection

Creating a user profile:

To Create a new user profile, click the *Create New User* button, then type your name in the popup text box, then click the *OK* button.

Selecting a user profile

Click your profile name from the *users list*.

On the right-hand side, the latest reaction time the system has on record for the selected user profile is displayed.

(The default reaction time for new user is 200ms, although this is updated dynamically when the new user actively uses Brake Aid)

After selecting your user profile, click *NEXT* at the bottom right of the window.

Begin Session

The begin session tab shows whether there are any valid game sessions running on the system that Brake Aid can attach to.

Once a valid game session is detected, the name of the game, racetrack and the configuration of the racetrack (if applicable) is displayed in the *Begin Session* tab.

The user has the option of either **starting a new Brake Aid Session** (which will require an initial benchmark lap to be driven before dynamic brake markers can be provided), or if the selected user has already used Brake Aid on the racetrack of the detected game before, they **can opt to Resume that last session**, and carry on with the dynamic brake markers that were saved at the end of that last session on that racetrack.

Using Brake Aid

Once a Brake Aid session is launched, the UI window will minimise to the system tray. To recall the UI during a session, right click the *Brake Aid icon* in the system tray and select *Show UI*.

Configuring the Brake Aid Overlay Light Graphic

Important: In order for the Brake Aid overlay graphic to be displayed over the game screen, as with most other third-party in-game telemetry overlays, the game itself must be run in either *Windowed Mode*, or *Full Screen Borderless Window Mode*.

In iRacing, *Full Screen Borderless Window Mode* is enabled from within a running iRacing sim session, by heading to:

***Settings* → *Display Settings* → *Display*, then toggling *Full Screen* to Off, and *Border* to Off.**

(The sim session may need to be restarted for the changes to take effect.)

Once a Brake Aid session is started/resumed, the overlay graphic will be displayed when the racing session itself is started (i.e. the game enters *driving mode*).

Once the overlay is displayed, it can be freely dragged anywhere on the screen and can be resized by either *dragging from the bottom right corner* of the overlay, or by *scrolling the mouse wheel* while the cursor is positioned over the overlay.

Exiting a Brake Aid Session

Once a Brake Aid session has begun, it will *automatically* exit (and save session data) once the game session it is attached to *completely* ends.

Alternatively, the session can be ended at any time from the Brake Aid UI (which can be displayed by right clicking the Brake Aid icon in the system tray and clicking *Show UI* and then clicking the *End Session* button in the *Begin Session* tab of *Brake Aid Setup*).

Button Mapping

Brake Aid currently has three user inputs, which include:

- Push Forward
- Pull Back
- Don't Count

Each of these functions can be mapped to either a keyboard key, or a button on a connected external peripheral, such as game controllers, racing sim wheels and other sim hardware that contains buttons.

Game Controller Compatibility

Most standard game devices that work on Windows should be compatible with Brake Aid.

Devices that need third-party drivers installed to operate correctly, may also be compatible with Brake Aid if they are listed within the Windows *Game Controllers* control panel applet, and the buttons are responsive within the *Test* tab of that device's *properties* menu, however this is ***not guaranteed***.

Regarding Keyboard Key Binding:

In addition to binding Brake Aid's user inputs to external peripheral buttons, alternatively, users also have the option to bind any of these inputs to keyboard keys.

Brake Aid assigns mapped inputs to keyboard keys by allocating each mapped key as a *hotkey* when a Brake Aid session is started/resumed.

During a Brake Aid session, any mapped keyboard keys are effectively *blocked* from the rest of the PC and are *only* operational with Brake Aid during this time.

When a Brake Aid session is ended, the mapped keyboard keys are deallocated from their hotkey state, and they become responsive to the rest of the PC again.

Only during an *active* Brake Aid session, are mapped keyboard keys blocked from the rest of the PC.

Mapping a Button/Key to a Brake Aid User Input Function:

On the left-hand side of the Brake Aid user interface, click *Button Mapping*.

To map one of the user input functions to a button or key, click the *MAP* button in that function's row.

If successful, the text under *Current Mapping* will be updated to confirm that the mapping was successful.

To remove a user input mapping, click *CLEAR MAPPING* in that function's row.

Brake Aid Configuration

The *configuration* page within the Brake Aid user interface contains various runtime variables that alter the behaviour of Brake Aid.

The default settings are optimised to allow Brake Aid to work effectively on a variety of racing setups.

If Brake Aid isn't working correctly or effectively for you, please reach out to the support team, who can assist with tweaking the configuration options to better suit your particular platform.

Feel free to adjust the Brake Aid configuration parameters if you want to experiment.

If you want to roll back to the default configuration options, click *Load Defaults* at the bottom right corner of the configuration page.

Troubleshooting

As Brake Aid is currently beta software, there is a high likelihood that operational issues may arise, and bugs may be discovered.

If the software crashes

The beta version of the Brake Aid software has extensive internal logging built in, which logs technical warnings and errors encountered in each of the subsystems of the software as it is running.

This beta version saves the complete set of logs for *up to 10 consecutive runs* of the Brake Aid software.

The most effective way for the support team to assist you with software crashes is to package the complete set of logs and email them to support.

This is easily achieved by double clicking the batch script ***Dump Brake Aid Logs to zip folder.bat*** which is located within the root folder of the Brake Aid software.

Once this script runs, a file called ***LogOutput.zip*** appears in the root folder. It is this zip file that should be sent to support, along with a brief description of the circumstances surrounding the software crash.

Reporting Bugs

There are likely to be many bugs and glitches that occur with this beta version of the Brake Aid software!

When a bug or glitch is encountered, please make a note of it and post it on the Brake Aid Beta Testing Discord page.

Third Party Notices

This software uses third-party components.

For acknowledgements and license information, see *Third Party Licenses/ThirdPartyNotices.txt* included with this software.